

| 1(a)                        | <table><tr><th rowspan="2">organelle</th><th colspan="3">cell type</th></tr><tr><th>animal cells</th><th>plant cells</th><th>prokaryotic cells</th></tr><tr><td>nucleus</td><td>✓</td><td>✓</td><td>✗</td></tr><tr><td>large permanent vacuole</td><td>✗</td><td>✓</td><td>✗ ;</td></tr><tr><td>rough endoplasmic reticulum</td><td>✓</td><td>✓</td><td>✗ ;</td></tr><tr><td>Golgi body</td><td>✓</td><td>✓</td><td>✗ ;</td></tr><tr><td>centrioles</td><td>✓</td><td>✗</td><td>✗ ;</td></tr></table> | organelle   | cell type         |  |  | animal cells | plant cells | prokaryotic cells | nucleus | ✓ | ✓ | ✗ | large permanent vacuole | ✗ | ✓ | ✗ ; | rough endoplasmic reticulum | ✓ | ✓ | ✗ ; | Golgi body | ✓ | ✓ | ✗ ; | centrioles | ✓ | ✗ | ✗ ; | 4 |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------|--|--|--------------|-------------|-------------------|---------|---|---|---|-------------------------|---|---|-----|-----------------------------|---|---|-----|------------|---|---|-----|------------|---|---|-----|---|
| organelle                   | cell type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |                   |  |  |              |             |                   |         |   |   |   |                         |   |   |     |                             |   |   |     |            |   |   |     |            |   |   |     |   |
|                             | animal cells                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | plant cells | prokaryotic cells |  |  |              |             |                   |         |   |   |   |                         |   |   |     |                             |   |   |     |            |   |   |     |            |   |   |     |   |
| nucleus                     | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ✓           | ✗                 |  |  |              |             |                   |         |   |   |   |                         |   |   |     |                             |   |   |     |            |   |   |     |            |   |   |     |   |
| large permanent vacuole     | ✗                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ✓           | ✗ ;               |  |  |              |             |                   |         |   |   |   |                         |   |   |     |                             |   |   |     |            |   |   |     |            |   |   |     |   |
| rough endoplasmic reticulum | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ✓           | ✗ ;               |  |  |              |             |                   |         |   |   |   |                         |   |   |     |                             |   |   |     |            |   |   |     |            |   |   |     |   |
| Golgi body                  | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ✓           | ✗ ;               |  |  |              |             |                   |         |   |   |   |                         |   |   |     |                             |   |   |     |            |   |   |     |            |   |   |     |   |
| centrioles                  | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ✗           | ✗ ;               |  |  |              |             |                   |         |   |   |   |                         |   |   |     |                             |   |   |     |            |   |   |     |            |   |   |     |   |
| 1(b)(i)                     | microvilli ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1           |                   |  |  |              |             |                   |         |   |   |   |                         |   |   |     |                             |   |   |     |            |   |   |     |            |   |   |     |   |
| 1(b)(ii)                    | <p>site of aerobic respiration, so produces / provides, ATP ;<br/>A provides energy<br/>R produces energy</p> <p>use of ATP in context of absorption of digested food for, active transport / (idea of) absorption against a concentration gradient<br/>or<br/>endocytosis / pinocytosis; A bulk transport into cell(s) ;</p>                                                                                                                                                                         | 2           |                   |  |  |              |             |                   |         |   |   |   |                         |   |   |     |                             |   |   |     |            |   |   |     |            |   |   |     |   |